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CSD 380 DevOps

The Technology Value Stream



What Is the Technology Value Stream?

- A value stream is the full flow of work needed to deliver value to the customer.
- In software, this includes planning, coding, testing, security review, release, deployment, and feedback.
- The technology value stream is often harder to see than other types of work because much of the work happens inside tools, queues, approvals, and handoffs.
- DevOps focuses on making this work visible and reducing delays between idea, code, deployment, and feedback.

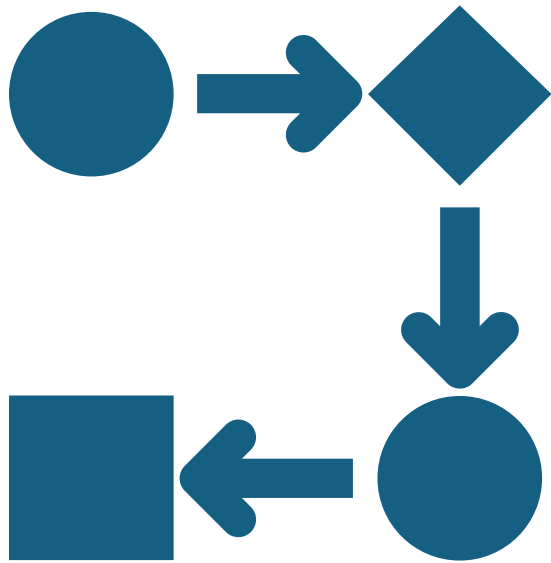


Lead Time vs. Processing Time

- Lead time: the total time from when a request is made until it is completed and delivered.
- Processing time: the time when work is being done on the request.
- The difference matters because long lead times are often caused by waiting, queues, approvals, rework, or handoffs, rather than by the actual hands-on work.
- For example, a change may take 2 hours to code and test, but sit in approval and release queues for 3 weeks.

The Common Issue: Deployment Lead Times of Months

- Many organizations used to have deployment lead times measured in weeks or months.
- Developers could complete code, but it waits for integration, testing, change approval, release windows, or handoffs.
- Large batch releases increased the risk because many changes were deployed together.



Why Long Deployment Lead Times Happen

- Work waits in queues between teams instead of flowing continuously.
- Manual testing, manual deployments, and manual approvals slow the process.
- Teams may optimize their own department rather than the entire value stream.
- Large changes are harder to work with.
- Poor visibility makes bottlenecks hard to identify and fix.



Deployment Lead Times in Minutes

- The DevOps ideal is to move changes from version control to production safely in minutes, not months.
- This requires small batch sizes, auto testing, continuous integration, auto deployment, and fast feedback.
- Shorter times help teams respond faster to customer requests, defects, security issues, and business needs.
- Speed must be balanced with stability by tracking quality and recovery metrics.

Traditional Flow vs. DevOps Flow

Traditional / Slow Flow

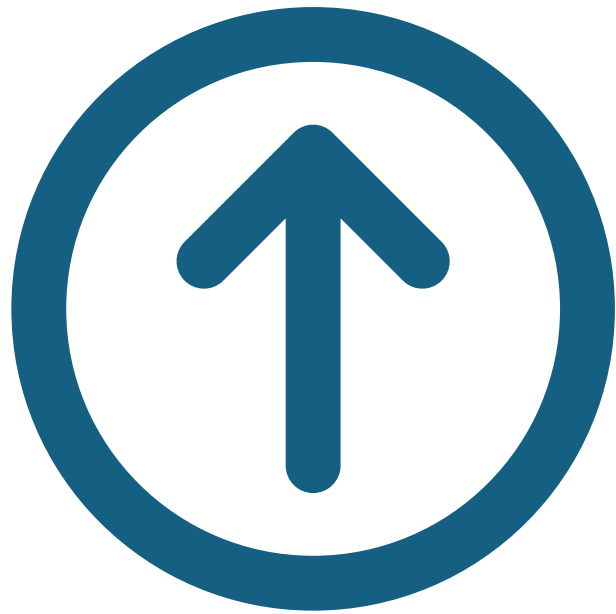
Lead Time: weeks or months



DevOps / Fast Flow

Lead Time: minutes or hours





How Teams Move Toward the DevOps Ideal

- Map the current value stream to find delays, repeated handoffs, and rework.
- Use small changes so each release is easier to test and troubleshoot.
- Automate builds, tests, security checks, and deployments where possible.
- Use continuous integration and deployment pipelines to create faster feedback.

Key Takeaways

- Lead time measures the total customer wait time; processing time measures the active work time.
- Long deployment lead times usually come from handoffs, manual steps, and large batches.
- The DevOps goal is not just speed; it is fast, safe, and repeatable delivery.
- A strong value stream makes work visible, removes bottlenecks, automates repeatable tasks, and improves feedback.



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